

12 (long)

A beam is designed to be as light as possible for the chassis of a race car. The beam is simply supported, carries a distributed load w (we can neglect self-weight here), and has a length L . The cross-sectional area A is uniform along the beam but its value may be varied. The deflection δ and the bending moment M_0 at mid-span are:

$$\delta = \frac{5wL^4}{384EI} \quad \text{and} \quad M_0 = \frac{wL^2}{8},$$

where E is the Young's modulus and I is the second moment of area. The maximum tensile stress

$$\sigma_0 = \frac{M_0 y_{\max}}{I}$$

is found at the maximum distance $y_{\max}$ of the beam cross-section from the neutral axis.

(a) In a first design iteration, the section shape of the beam may be changed.

(i) Using a solid square section as reference shape, derive the following expressions ϕ_e and ϕ_f of the shape factors for bending stiffness and strength,

$$\phi_e = \frac{12I}{A^2} \quad \text{and} \quad \phi_f = \frac{6(I/y_{\max})}{A^{3/2}},$$

respectively. [3]

(ii) There are two possible design constraints to consider: the beam deflection should not exceed $\delta_{\max}$, and the beam should not fail, $\sigma_0 \leq \sigma_f$ (the failure strength).

For each constraint, derive the performance index to be maximised for minimum weight. [7]

(iii) For each constraint, rank the materials given in Table 1 from best to worst. [4]

(b) In a refined model, the beam is a cylindrical tube with length $L = 1$ m, radius $R = 2$ cm and a thickness t that can be chosen. The distributed load $w = 10^3 \text{ N m}^{-1}$ and the maximum deflection $\delta_{\max} = 2.5$ mm. The tube is thin walled, $t \ll R$, so we may use $A = 2\pi R t$ and $I = \pi R^3 t$. Both stiffness and strength constraints identified in (a.ii) must now be satisfied.

(i) State the objective, geometric constraints, functional constraints, and the free variables for this design. [3]

(ii) Calculate the required value of the thickness t for the materials in Table 1. [7]

(iii) For each material, and using the values of t found in (b.ii), evaluate the relevant shape factor for the tube. Identify the lightest material for the beam, briefly noting other factors to consider. [6]

Material	Density ρ (Mg m ⁻³)	Modulus E (GPa)	Strength σ_f (MPa)	Maximum shape factor ϕ_e (ref. solid square)	Maximum shape factor ϕ_f (ref. solid square)
Steel	7.8	210	900	64	13
Al alloy	2.6	70	200	49	10
Ti alloy	4.5	100	700	57	11

Table 1

END OF PAPER